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Practical 8

Saving the JWT Token:

```
PS C:\Users\DELL\OneDrive\Documents\S DBE&DBD\Practical\student-records\student-records> $token = "eyJhbGciOiJIUzI1NiJ9.eyJzdWUiOiJwcm15YW5rYSIsImIhdCI6MTc5MDc..."
PS C:\Users\DELL\OneDrive\Documents\S DBE&DBD\Practical\student-records\student-records> $headers = @{
>>     Authorization = "Bearer $token"
>> }
```

Folder Structure:

Week-8_Practical_Student_Records/

```
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|   |   |   |   |   |   |   |   |
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|   |   |   |   |   |   |   |   |   |
|   |   |   |   |   |   |   |   |   | └─ dto/
|   |   |   |   |   |   |   |   |   |   |
|   |   |   |   |   |   |   |   |   |   | └─ entity/
|   |   |   |   |   |   |   |   |   |   |   |
|   |   |   |   |   |   |   |   |   |   |   | └─ exception/
|   |   |   |   |   |   |   |   |   |   |   |   |
|   |   |   |   |   |   |   |   |   |   |   |   | └─ repository/
```

```

| | | └─ security/
| | | └─ service/
| | |
| | └─ resources/
| | └─ application.properties
| |
| └─ target/ Codes:

```

1. Main Application

□ StudentRecordsApplication.java

```

package com.example.student_records; import
org.springframework.boot.SpringApplication; import
org.springframework.boot.autoconfigure.SpringBootApplication;
@SpringBootApplication public class
StudentRecordsApplication {
    public static void main(String[] args)
    {
        SpringApplication.run(StudentRecordsApplication.class,
args);
    }
}

```

2. Entity

□ Student.java

```

package com.example.student_records.entity;
import org.springframework.data.annotation.Id; import
org.springframework.data.mongodb.core.mapping.Document;
@Document(collection="students")
public class Student {
    @Id private
String id; private
String name;
private Integer age;
private String course;
private String email;
public String getId() {
return id; } public
void setId(String id) {

```

```

this.id = id;          }
public String getName() {
return name;
    }      public void setName(String
name) {          this.name = name;      }
public Integer getAge() {          return
age;
    }      public void
setAge(Integer      age)      {
this.age = age;          }
public String getCourse() {
return course;
    }      public void setCourse(String
course) {          this.course = course;
    }      public String
getEmail() {          return email;
    }      public void setEmail(String
email) {          this.email = email;
    }
}

```

□ User.java

```

package com.example.student_records.entity; import
org.springframework.data.annotation.Id; import
org.springframework.data.mongodb.core.index.Indexed; import
org.springframework.data.mongodb.core.mapping.Document;
@Document(collection = "users")
public class User {
    @Id      private
String id;
    @Indexed(unique = true)
private String username;
private String password;
    public String getId() {
return id;
    }      public void setId(String id) {
this.id = id;          }      public String
getUsername() {          return username;      }
public void setUsername(String username) {
this.username = username;      }      public
String getPassword() {          return password;
    }      public void setPassword(String password)
{          this.password = password;
    }
}

```

3. DTO

□ StudentRequest.java

```
package com.example.student_records.dto;
import jakarta.validation.constraints.Email;
import jakarta.validation.constraints.Min;
import jakarta.validation.constraints.NotBlank;
import jakarta.validation.constraints.NotNull;
public class StudentRequest
{
    @NotBlank(message = "name is required")    private
    String name;

    @NotNull(message = "age is required")
    @Min(value = 16, message = "age must be at least 16")
    private Integer age;

    @NotBlank(message = "course is required")
    private String course;

    @NotBlank(message = "email is required")
    @Email(message = "a valid email is required")
    private String email;    public String
    getName() {    return name;
    }    public void setName(String
    name) {    this.name = name;
    }    public Integer
    getAge() {
    return age;    }    public
    void setAge(Integer age) {
    this.age = age;    }
    public String getCourse() {
    return course;
    }    public void setCourse(String
    course) {    this.course = course;
    }    public String
    getEmail() {    return
    email;
    }    public void setEmail(String
    email) {    this.email = email;
    }
}
```

□ RegisterRequest.java

```

package com.example.student_records.dto;
import
jakarta.validation.constraints.NotBlank; import
jakarta.validation.constraints.Size;
public class RegisterRequest
{

    @NotBlank    private
String username;

    @NotBlank
    @Size(min = 6, message = "password must be at least 6 characters")
private String password;
    public String getUsername()
{
    return username;
}
    public void setUsername(String
username) {
    this.username = username;
}
    public String
getPassword() {
    return
password;
}
    public void setPassword(String
password) {
    this.password = password;
}
}

```

□ LoginRequest.java package com.example.student_records.dto;

```

import jakarta.validation.constraints.NotBlank

;

public class LoginRequest
{

    @NotBlank    private
String username;

    @NotBlank    private String
password;
    public String
getUsername() {
    return
username;
}
    public void setUsername(String
username) {
    this.username = username;
}
    public String
getPassword() {
    return
password;
}
    public void setPassword(String
password) {
    this.password = password;
}
}

```

```
}
```

4. Repository

- StudentRepository.java

```
package com.example.student_records.repository;
import com.example.student_records.entity.Student;
import
org.springframework.data.mongodb.repository.MongoRepository;
public interface StudentRepository extends MongoRepository<Student,
String>
{
}
```

- UserRepository.java

```
package com.example.student_records.repository;
import
com.example.student_records.entity.User;
import org.springframework.data.mongodb.repository.MongoRepository;
import
java.util.Optional; public
interface UserRepository
extends
MongoRepository<User, String>
{
    Optional<User> findByUsername(String username);
}
```

5. Service

- StudentService.java

```
package com.example.student_records.service; import
com.example.student_records.dto.StudentRequest; import
com.example.student_records.entity.Student; import
com.example.student_records.repository.StudentRepository; import
org.springframework.stereotype.Service;
import java.util.List;
import java.util.Optional;
@Service public class
StudentService { private
final StudentRepository
repository;
    public StudentService(StudentRepository repository) {
this.repository = repository;
```

```

    }    public Student
create(StudentRequest r) {

    Student s = new Student();
    apply(s,
r);

    return
repository.save(s);    }
public List<Student> getAll() {
return repository.findAll();
}

    public Optional<Student> getById(String id) {
return repository.findById(id);
} public Optional<Student> update(String id, StudentRequest

r) {    return repository.findById(id).map(s -> {

apply(s, r);
return
repository.save(s);
});
    }    public boolean delete(String
id) {        if
(!repository.existsById(id)) {
return false;        }
repository.deleteById(id);
return true;
    }    private void apply(Student s,
StudentRequest r) {

    s.setName(r.getName());
    s.setAge(r.getAge());
    s.setCourse(r.getCourse());
    s.setEmail(r.getEmail());
    }
}

```

6. Controller

□ StudentController.java

```

package com.example.student_records.controller;
import com.example.student_records.dto.StudentRequest;
import com.example.student_records.entity.Student; import
com.example.student_records.service.StudentService;
import jakarta.validation.Valid; import
org.springframework.http.ResponseEntity; import

```

```

org.springframework.web.bind.annotation.*;
import java.util.List;
@RestController
@RequestMapping("/students") public class
StudentController {      private final
StudentService service;      public
StudentController(StudentService service)
{      this.service = service;
    }

    @PostMapping      public
    ResponseEntity<Student> create(
        @Valid @RequestBody StudentRequest r) {
    return
    ResponseEntity
        .status(201)
        .body(service.create(r));
    }

    @GetMapping      public
    List<Student> getAll() {      return
    service.getAll();
    }

    @GetMapping("/{id}")      public
    ResponseEntity<Student> getById(
    @PathVariable String id) {
    return service.getById(id)
        .map(ResponseEntity::ok)
        .orElse(ResponseEntity.notFound().build());
    }

    @PutMapping("/{id}")
    public ResponseEntity<Student> update(
        @PathVariable String id,
        @Valid @RequestBody StudentRequest r) {
    return service.update(id, r)
        .map(ResponseEntity::ok)
        .orElse(ResponseEntity.notFound().build());
    }

    @DeleteMapping("/{id}")      public
    ResponseEntity<Void> delete(
    @PathVariable String id) {

        return service.delete(id)
            ? ResponseEntity.noContent().build()
            : ResponseEntity.notFound().build();
    }
}

```


□ AuthController.java

```
package com.example.student_records.controller; import
com.example.student_records.dto.LoginRequest; import
com.example.student_records.dto.RegisterRequest; import
com.example.student_records.entity.User; import
com.example.student_records.repository.UserRepository; import
com.example.student_records.security.JwtUtil;
import
jakarta.validation.Valid;
import
org.springframework.http.ResponseEntity;
import org.springframework.security.authentication.AuthenticationManager;
import
org.springframework.security.authentication.UsernamePasswordAuthentication
T oken; import
org.springframework.security.crypto.password.PasswordEncoder;
import
org.springframework.web.bind.annotation.*;
import
java.util.Map;

@RestController
@RequestMapping("/auth")
public class AuthController {
    private final UserRepository users;
    private final PasswordEncoder encoder; private
    final AuthenticationManager manager; private
    final JwtUtil jwt;
    public
    AuthController(
        UserRepository users,
            PasswordEncoder encoder,
            AuthenticationManager manager,
            JwtUtil jwt) {
        this.users =
users;
        this.encoder = encoder;
        this.manager = manager; this.jwt
        = jwt;
    }

    @PostMapping("/register")
    public ResponseEntity<?> register(
        @Valid @RequestBody RegisterRequest r) { if

        (users.findByUsername(r.getUsername()).isPresent()) {
```

```

        return ResponseEntity.badRequest()
            .body(Map.of(
                "error",
                "username already taken"
            ));
    }

    User user = new User();

    user.setUsername(r.getUsername());
    user.setPassword(
        encoder.encode(r.getPassword())
    );
    users.save(user);
    return
    ResponseEntity.status(201)
        .body(Map.of(
            "message",
            "registered successfully"
        ));
    }

    @PostMapping("/login")    public
    ResponseEntity<?> login(
        @Valid @RequestBody LoginRequest r) {
        manager.authenticate(
            new
            UsernamePasswordAuthenticationToken(
                r.getUsername(),
                r.getPassword()
            )
        );

        return ResponseEntity.ok(
            Map.of(
                "token",
                jwt.generateToken(r.getUsername())
            )
        );
    }
}

```

7. Security

□ SecurityConfig.java


```

        .authorizeHttpRequests(auth ->
auth
            .requestMatchers("/auth/**")
            .permitAll()
            .anyRequest()
            .authenticated()
        )

        .addFilterBefore(
jwtAuthFilter,
            UsernamePasswordAuthenticationFilter.class
        )

        .build();
    }
}

```

□ JwtUtil.java

```

package com.example.student_records.security;
import io.jsonwebtoken.*;
import
io.jsonwebtoken.security.Keys;
import
org.springframework.beans.factory.annotation.Value; import
org.springframework.stereotype.Component;
import javax.crypto.SecretKey; import
java.nio.charset.StandardCharsets;
import java.util.Date;
@Component public class JwtUtil {

    @Value("${jwt.secret}")    private
String secret;

    @Value("${jwt.expiration-ms}")    private
long expirationMs;
    private SecretKey key()
{
        return
Keys.hmacShaKeyFor(
secret.getBytes(StandardCharsets.UTF_8)
        );
    }
    public String generateToken(String username)
{

        Date now = new Date();
        return Jwts.builder()
.setSubject(username)
        .setIssuedAt(now)

```

```

        .setExpiration(new Date(now.getTime()
+ expirationMs)
        )
        .signWith(key(), SignatureAlgorithm.HS256)
        .compact();
    }
    public String extractUsername(String
token) {
        return Jwts.parserBuilder()
        .setSigningKey(key())
        .build()
        .parseClaimsJws(token)
        .getBody()
        .getSubject();
    }
    public boolean isValid(String token) {
    try {

        Jwts.parserBuilder()
        .setSigningKey(key())
        .build()
        .parseClaimsJws(token);

        return true;

    } catch (JwtException | IllegalArgumentException e) {
return
false;
    }
    }
}

```

□ JwtAuthFilter.java

```

package com.example.student_records.security;
import jakarta.servlet.*; import
jakarta.servlet.http.*;
import
org.springframework.security.authentication.UsernamePasswordAuthentication
Token; import
org.springframework.security.core.context.SecurityContextHolder; import
org.springframework.security.core.userdetails.UserDetailsService; import
org.springframework.security.web.authentication.WebAuthenticationDetailsSo
urce;
import org.springframework.stereotype.Component; import
org.springframework.web.filter.OncePerRequestFilter;
import
java.io.IOException;

@Component public class JwtAuthFilter extends
OncePerRequestFilter {

```

```

        private final JwtUtil jwtUtil;        private final
        UserDetailsService userDetailsService;
        public
        JwtAuthFilter(
            JwtUtil jwtUtil,
            UserDetailsService userDetailsService) {
this.jwtUtil =
jwtUtil; this.userDetailsService
= userDetailsService;    }

    @Override
    protected void doFilterInternal(
        HttpServletRequest request,
        HttpServletResponse response,
        FilterChain chain) throws
        ServletException, IOException {
        String header = request.getHeader("Authorization");
        if (header != null && header.startsWith("Bearer ")) {
            String token = header.substring(7);
            if (jwtUtil.isValid(token))
            {

                System.out.println("JWT VALID");
                String username =
                jwtUtil.extractUsername(token);
                var details =
                userDetailsService.loadUserByUsername(username);
                var authentication =
                new UsernamePasswordAuthenticationToken(
                details,
                null,
                details.getAuthorities()
                );

                authentication.setDetails(
                    new WebAuthenticationDetailsSource()
                        .buildDetails(request)
                );

                SecurityContextHolder
                    .getContext()
                    .setAuthentication(authentication);
            } else {

                System.out.println("JWT INVALID");
            }
        }
        chain.doFilter(request, response);
    }
}

```

□ CustomUserDetailsService.java

```
package com.example.student_records.security; import
com.example.student_records.repository.UserRepository;
import
org.springframework.security.core.userdetails.*; import
org.springframework.stereotype.Service;
@Service public class CustomUserDetailsService implements
UserDetailsService { private final UserRepository
userRepository;
    public CustomUserDetailsService(UserRepository userRepository)
{
    this.userRepository = userRepository;
}

@Override public UserDetails
loadUserByUsername(String username) throws
UsernameNotFoundException { var user =
userRepository
    .findByUsername(username)
    .orElseThrow(() -> new
UsernameNotFoundException("User not found"));
    return
User.withUsername(user.getUsername())
        .password(user.getPassword())
        .authorities("USER")
        .build();
}
}
```

8. Exception Handling

□ GlobalExceptionHandler.java

```
package com.example.student_records.exception;
import org.springframework.http.ResponseEntity;
import
org.springframework.web.bind.MethodArgumentNotValidException; import
org.springframework.web.bind.annotation.ExceptionHandler; import
org.springframework.web.bind.annotation.RestControllerAdvice;
import
java.util.HashMap; import
java.util.Map;
@RestControllerAdvice public class
GlobalExceptionHandler {

    @ExceptionHandler(MethodArgumentNotValidException.class)
    public ResponseEntity<Map<String, Object>> handleValidation(
        MethodArgumentNotValidException ex) {
```

```

        Map<String, String> errors = new HashMap<>();
ex.getBindingResult()
    .getFieldErrors()                .forEach(error ->
errors.put(
    error.getField(),
    error.getDefaultMessage()
)
    );

    Map<String, Object> body = new HashMap<>();
body.put("error", "VALIDATION_ERROR");
body.put("details", errors);
    return
ResponseEntity.badRequest().body(body);
    }
}

```

9. Configuration

```

application.properties

spring.application.name=student-records
spring.data.mongodb.uri=mongodb://localhost:27017/student_records
jwt.secret=replace-this-with-a-long-random-secret-at-least-32-characters
jwt.expiration-ms=3600000

server.port=8080

```

10. Maven Configuration

```

❏ pom.xml

<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <parent>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-parent</artifactId>
        <version>3.5.6</version>
        <relativePath/>
    </parent>

```



```
<groupId>com.example</groupId>
<artifactId>student-records</artifactId>
<version>0.0.1-SNAPSHOT</version>
<name>student-records</name>
<description>Student Records Application</description>
<properties>
    <java.version>17</java.version>
</properties>
<dependencies>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-web</artifactId>
    </dependency>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-data-mongodb</artifactId>
    </dependency>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-validation</artifactId>
    </dependency>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-security</artifactId>
    </dependency>
    <dependency>
        <groupId>io.jsonwebtoken</groupId>
        <artifactId>jjwt-api</artifactId>
        <version>0.11.5</version>
    </dependency>
    <dependency>
        <groupId>io.jsonwebtoken</groupId>
        <artifactId>jjwt-impl</artifactId>
        <version>0.11.5</version>
        <scope>runtime</scope>
    </dependency>
    <dependency>
        <groupId>io.jsonwebtoken</groupId>
        <artifactId>jjwt-jackson</artifactId>
        <version>0.11.5</version>
        <scope>runtime</scope>
    </dependency>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-test</artifactId>
        <scope>test</scope>
    </dependency>
</dependencies>
```

```
<build>
  <plugins>
    <plugin>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-maven-plugin</artifactId>
    </plugin>
  </plugins>
</build>
```

</project>

Spring Boot Application Running Successfully

```
2026-09-30T18:42:32.479+05:30 INFO 12212 --- [student-records]
[main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tom
cat started on port 8080 (http) with context path '/'
2026-09-30T18:42:32.486+05:30 INFO 12212 --- [student-records]
[main] c.e.s.StudentRecordsApplication : Sta
rted StudentRecordsApplication in 4.066 seconds (process running
for 4.735)
```